📘 Technical Manual: Hybrid Adaptive Pneumatic Control System

1. System Architecture

This system utilizes a **Hybrid Adaptive Control Strategy** implemented via a Finite State Machine (FSM). It addresses the non-linear pneumatic characteristics of soft actuators (compressibility, hysteresis, and strain hardening) by switching between two distinct control regimes:

1. Coarse Control Regime (Continuous Flow):

When the error magnitude exceeds the COARSE_THRESHOLD, the system engages the pumps continuously. This maximizes the slew rate to reach the target vicinity quickly, minimizing rise time.

2. Fine Control Regime (Discrete Pulse Modulation):

When the error is within the COARSE_THRESHOLD, the system switches to a discrete control loop. It calculates a specific actuation duration (t_{pulse}) based on a Gain Scheduled Proportional Controller. This compensates for the variable stiffness of the actuator at different pressure levels.

3. Relaxation State (Pneumatic Settling):

Between actuation events, the system enters a RESTING state. This open-loop pause allows dynamic pressure transients ("ringing") to decay, ensuring the subsequent sensor measurement reflects the true static pressure of the system.

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2. Control Parameters & Tuning Guide

The following parameters define the system's transfer function. Tuning involves adjusting the gains and thresholds to optimize the transient response (speed) and steady-state accuracy.

A. System Constants (Deadband & Target)

Parameter	Default	Technical Description	Tuning Implication
TARGET_P	10.0	**Setpoint:** The desired pressure state (kPa).	Dynamically updated via Serial Interface.
E	0.5	**Error Deadband:** The hysteresis range ($\pm E$) around the setpoint where no control action is taken.	**Increase:** Improves stability but reduces precision.

Parameter	Default	Technical Description	Tuning Implication
			Decrease: Increases precision but may induce limit-cycle oscillations (hunting).

B. Regime Switching (Coarse Tuning)

These parameters govern the transition between the continuous and discrete control modes.

Parameter	Default	Technical Description	Effect of INCREASING
COARSE_THRESHOLD	3.0	**Switching Threshold:** The error magnitude defining the boundary between Coarse and Fine regimes.	**Effect:** Extends the continuous flow phase. **Risk:** If too large, the high-inertia flow may cause overshoot before the controller can switch to fine mode.
COARSE_STOP	1.0	**Predictive Offset:** A negative bias applied during Coarse mode to compensate for dynamic pressure spikes (sensor lead).	**Increase:** Terminates Coarse mode earlier (Undershoot). **Decrease:** Terminates Coarse mode later (closer to setpoint).

C. Adaptive Gain Scheduling (Fine Tuning)

The pulse duration is calculated using a Linear Parameter-Varying (LPV) equation:

$$t_{pulse} = |e(t)| \times (K_{base} + K_{slope} \cdot P_{current})$$

Parameter	Default	Technical Description	Effect of INCREASING
Kp_Inflate_Base	20.0	**Base Proportional Gain (K_p):** The linear gain coefficient applied to the error.	**Effect:** Increases system aggression globally. Increases pulse width for a given error.

Parameter	Default	Technical Description	Effect of INCREASING
Kp_Inflate_Slope	1.0	**Scheduling Gain (K_{slope}):** Adds a pressure-dependent gain component to compensate for increased pneumatic resistance at higher pressures.	**Effect:** Increases actuation strength primarily at high static pressures. Lineariuzes the system response.
MIN_PULSE	50	**Minimum Actuation Time:** The lower bound for actuator activation, overcoming valve stiction and electromechanical latency.	**Effect:** Forces a minimum quantization step. Prevents "stalling" near the setpoint due to ineffective micro-pulses.

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3. System Characterization & Tuning Strategy

Analyze the step response of the system to diagnose and correct control behaviors.

Scenario 1: Steady-State Error (Stalling)

- **Observation:** The system stabilizes outside the deadband (e.g., $P_{actual} = 9.5$, Target = 10.0, $E = 0.5$). The calculated pulse is insufficient to overcome pneumatic resistance.
- **Corrective Action:**
    1. **Increase MIN_PULSE:** Ensures the minimum control effort is sufficient to produce a measurable pressure change.
    2. **Increase Kp_Inflate_Base:** Increases the gain to generate longer pulses for small errors.

Scenario 2: Limit Cycle Oscillation (Overshoot)

- **Observation:** The system oscillates around the setpoint, repeatedly overshooting and venting. The loop gain is too high.
- **Corrective Action:**
    1. **Decrease Kp_Inflate_Base:** Reduces the proportional gain to dampen the response.
    2. **Increase REST_TIME:** Allows for full dissipation of dynamic pressure waves, preventing aliased sensor readings from triggering false corrections.

Scenario 3: Slow Transient Response (Under-damped Coarse Mode)

- **Observation:** The system exits Coarse mode too early, resulting in a long duration of fine pulsing to reach the target.
- **Corrective Action:**
    1. **Decrease COARSE_THRESHOLD:** Keeps the system in the high-flow continuous regime closer to the target.
    2. **Decrease COARSE_STOP:** Reduces the predictive braking offset, allowing the coarse phase to terminate closer to the setpoint.

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4. Operational Protocols

1. **Initialization:**
    - Ensure pneumatic lines are depressurized before booting.
    - Verify the baud rate is set to **115200**.
2. **Sensor Zeroing:**
    - Observe the pressure reading at atmospheric conditions.
    - If the reading deviates from 0.00 kPa, adjust the SENSOR_OFFSET constant in the firmware to calibrate the sensor baseline.
3. **Command Interface:**
    - Send floating-point values via Serial (e.g., 25.0) to update the reference setpoint TARGET_P.
    - Monitor the [STATUS] log for real-time state machine feedback.

5. Safety & Signal Processing

- **Signal Conditioning:** The readSensor() function applies a linear transformation to the ADC voltage, clamped to the sensor's valid range ($0.5V - 4.5V$).
- **Fault Detection:**
    - **Open Circuit:** Voltage $< 0.2V$ triggers an immediate system halt (stopAll()) and error logging.
    - **Over-Range:** Voltage $> 4.5V$ triggers safety shutdown.
- **Temporal Filtering:** The REST_TIME delay acts as a mechanical low-pass filter, preventing the controller from reacting to high-frequency pneumatic noise.